

$$1A) D_4 = \{ \langle x, y \rangle \mid x^4 = y^2 = e \text{ ve } xy = yx^3 \}$$

$$= \{ e, x, x^2, x^3, y, xy, x^2y, x^3y \} \quad (1)$$

$$o(D_4) = 8$$

$$o(e) = 2^0 = 1$$

$$o(x) = 2^2 = 4$$

$$o(x^2) = 2^1 = 2$$

$$o(x^3) = 2^2 = 4$$

$$o(y) = 2^1 = 2$$

$$o(xy) = 2^1 = 2 \quad (4)$$

$$o(x^2y) = 2^1 = 2$$

$$o(x^3y) = 2^1 = 2$$

$p = 2$ gruptur.

Mathe Hexaplam:

$$xyxy = yx^3xy = yx^4y = y^2 = e \quad (5)$$

$$x^2yx^2y = xxyx^2y = xyx^3xy = xyxy = e$$

$$x^3yx^3y = x^2xyx^3y = x^2yx^6y = x^2yx^2y = xxyx^4y = xxyx^2y = xyx^5y = xyxy = e$$

$$1B) D_6 = \{ \langle x, y \rangle \mid x^6 = y^2 = e \text{ ve } xy = yx^5 \}$$

$$= \{ e, x, x^2, x^3, x^4, x^5, y, xy, x^2y, x^3y, x^4y, x^5y \}$$

$$o(D_6) = 12 \quad (2)$$

$$C_6 = \{ e, x, x^2, x^3, x^4, x^5 \}, \quad o(C_6) = 6$$

$$[D_6 : C_6] = \frac{o(D_6)}{o(C_6)} = \frac{12}{6} = 2 \text{ koset} \quad (2)$$

$$C_6 = \{ e, x, x^2, x^3, x^4, x^5 \} \quad b = \text{koset} \quad (2)$$

$$yC_6 = \{ y, yx, yx^2, yx^3, yx^4, yx^5 \} \quad \text{sol-koset} \quad (2)$$

$$C_6y = \{ y, xy, x^2y, x^3y, x^4y, x^5y \} \quad \text{sag-koset}$$

$$\begin{array}{cccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ y & yx^5 & yx^4 & yx^3 & yx^2 & yx \end{array}$$

$$C_6 \triangleleft D_6 \quad \text{ve} \quad D_6 = C_6 \cup yC_6 = C_6 \cup C_6y$$

$$D_6/C_6 = \{ C_6, yC_6 \} = \{ C_6, C_6y \} \text{ dir.} \quad (2)$$

2A) $(\mathbb{Z}_8^*, \cdot) \cong V_4 = \text{Klein-4}$

$\mathbb{Z}_8^* = \{x \mid (x, 8) = 1\} = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$ (1)

$o(\mathbb{Z}_8^*) = 4$

$\langle \bar{1} \rangle = \{\bar{1}\}$

$\langle \bar{3} \rangle = \{\bar{1}, \bar{3}\}$

$\langle \bar{5} \rangle = \{\bar{1}, \bar{5}\}$

$\langle \bar{7} \rangle = \{\bar{1}, \bar{7}\}$

$o(1) = 1$

$o(3) = 2$

$o(5) = 2$

$o(7) = 2$

Üreter yok. Devirli değildir.

$V_4 = \text{Klein-4}$

$V_4 = \{e, a, b, c \mid a^2 = b^2 = c^2 = e, ab = c, ac = b, bc = a\}$ (1)

$\mathbb{Z}_8^*$ devirli grup değildir. Elementler 1, 2, 2, 2 mertebeli.

V_4 devirli grup değildir. Elementler 1, 2, 2, 2 mertebeli.

$o(e) = 1$

$o(a) = 2$

$o(b) = 2$

$o(c) = 2$

$\langle e \rangle = \{e\}$

$\langle a \rangle = \{e, a\}$

$\langle b \rangle = \{e, b\}$

$\langle c \rangle = \{e, c\}$

(3)

Üreter yok. Devirli değildir.

4 new mertebeden devirli olmayan her grup $G \cong V_4$ dir.

$1 \rightarrow e$

$3 \rightarrow a$

$5 \rightarrow b$

$7 \rightarrow c$

$f: \mathbb{Z}_8^* \rightarrow V_4$
 $\mathbb{Z}_8^* \cong V_4$

2B) G Abel grup $n \in \mathbb{Z}^+$ olsun

$H = \{a \in G \mid x \in G \text{ için } a = x^n\}$, G new alt grubu mu?

$e \in G$, $x = e$, $x^n = e^n = e = a \in H \neq \emptyset$

$a \in H, b \in H \Rightarrow a = x^n \text{ ve } b = y^n$

$ab = x^n \cdot y^n = (xy)^n$, G Abel. (5)

$a \in H$, $a^{-1} \in H$? $a = x^n \Rightarrow a^{-1} = (x^n)^{-1}$ den.

$a^{-1} = (x)^{-n} = (x^{-1})^n \in H$ (5)

$\Rightarrow H < G$ dir

3A) Alt halka $(R, +, \cdot)$ bir halka olsun.

$\emptyset \neq R' \subset R$ olmak üzere $(R', +, \cdot)$ için

$(R', +)$ Abel grp. $< (R, +)$ ise

$(R', \cdot)$ Yarı-grp $< (R, \cdot)$

veya $\forall a, b \in R'$ için $a \cdot b \in R'$ ve $\forall a, b \in R'$ için $a \cdot b \in R'$
 $(R', +, \cdot)$ alt halka adını alır. (2)

İdeal: $(R, +, \cdot)$ bir halka ve $I \subseteq R$ alt küme

olsun, - $a, b \in I$ için $a \cdot b \in I$ (3)
 $a, b \in I$ için $a - b \in I$
 $r \in R, a \in I$ için $ra \in I$ veya $ar \in I$
 Sol ideal sağ-ideal

I 'ya R halkasının ideali denir.

Her ideal bir alt halkadır. Fakat tersi doğru değildir. (2)

Örneğin $(\mathbb{Z}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$ bir alt halkasıdır.

Fakat $1 \in \mathbb{Z}$ ve $\frac{1}{2} \in \mathbb{R}$ için (3) $1 \cdot \frac{1}{2} = \frac{1}{2} \notin \mathbb{Z}$ dir

$\mathbb{Z}$, $\mathbb{R}$ nin bir ideali değildir.

$$3B) I_2(\mathbb{Z}) = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}, I = \begin{bmatrix} 0 & b \\ 0 & c \end{bmatrix}$$

I , $I_2(\mathbb{Z})$ nin ideali midir?

$$i) \begin{bmatrix} 0 & b_1 \\ 0 & c_1 \end{bmatrix} - \begin{bmatrix} 0 & b_2 \\ 0 & c_2 \end{bmatrix} = \begin{bmatrix} 0 & b_1 - b_2 \\ 0 & c_1 - c_2 \end{bmatrix} \in I \quad (3)$$

$$ii) \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \cdot \begin{bmatrix} 0 & b_1 \\ 0 & c_1 \end{bmatrix} = \begin{bmatrix} 0 & ab_1 + bc_1 \\ 0 & cc_1 \end{bmatrix} \in I \quad (3)$$

$$\begin{bmatrix} 0 & b_1 \\ 0 & c_1 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} = \begin{bmatrix} 0 & b_1c \\ 0 & c_1c \end{bmatrix} \in I \quad (3)$$

$a \cdot r = ar \in I$ sağlanır.

D halde I , $I_2(\mathbb{Z})$ nin bir idealidir. (1)

4) $\mathbb{Z}_{15}$ i) Sylow p -alt grupları?

i) $\mathbb{Z}_{15} = \{0, 1, 2, 3, \dots, 13, 14\}$ $o(\mathbb{Z}_{15}) = 15 = 3 \cdot 5$

Sylow 3-alt grup ve Sylow 5-alt gruplar vardır.

$$G_3 = \{0, 5, 10\} = \langle 5 \rangle \quad (2)$$

$$G_5 = \{0, 3, 6, 9, 12\} = \langle 3 \rangle \quad (2)$$

$$\mathbb{Z}_{15} = G_3 \oplus G_5$$

ii) iç direkt çarpım olarak ifade et.

$$H_1 = \langle 5 \rangle, \quad H_2 = \langle 3 \rangle$$

$$H_1 + H_2 = \{h_1 + h_2 \mid h_1 \in \langle 5 \rangle \text{ ve } h_2 \in \langle 3 \rangle\} \quad (2)$$
$$= \mathbb{Z}_{15}$$

$$H_1 \cap H_2 = \langle 5 \rangle \cap \langle 3 \rangle = \{0\} = \{e_{\mathbb{Z}_{15}}\} \quad (1)$$

$$h_1 + h_2 = h_2 + h_1 \quad \text{mi?}$$

$$\bar{5} + \bar{3} = \bar{3} + \bar{5}, \quad \bar{5} + \bar{6} = \bar{6} + \bar{5} \quad \dots \quad (1)$$

$$\mathbb{Z}_{15} \cong \langle \bar{5} \rangle \oplus \langle \bar{3} \rangle \quad \text{dir.} \quad (1)$$

iii) Dış direkt çarpım

$$\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n, \quad (m, n) = 1 \text{ iken.}$$

$$(3, 5) = 1 \Rightarrow \mathbb{Z}_{15} \cong \mathbb{Z}_5 \times \mathbb{Z}_3 \quad (3) \text{ dir.}$$

iv) $\text{Oto}(\mathbb{Z}_{15}) = \{f_k \mid (k, 15) = 1\} \quad (2)$
 $= \{f_1, f_2, f_4, f_7, f_8, f_{11}, f_{13}, f_{14}\}$

$$f_1(0) = 1 \cdot 0 = 0$$

$$f_1(1) = 1 \cdot 1 = 1$$

$$f_1(2) = 1 \cdot 2 = 2$$

⋮

$$f_1(14) = 1 \cdot 14 = 14$$

$$f_2(0) = 2 \cdot 0 = 0$$

$$f_2(1) = 2 \cdot 1 = 2 \quad (2)$$

$$f_2(2) = 2 \cdot 2 = 4$$

$$f_2(3) = 2 \cdot 3 = 6$$

⋮

$$f_2(14) = 2 \cdot 14 = 13$$

$$f_4(0) = 4 \cdot 0 = 0$$

$$f_4(1) = 4 \cdot 1 = 4$$

$$f_4(2) = 4 \cdot 2 = 8$$

$$f_4(3) = 4 \cdot 3 = 12$$

$$f_4(4) = 4 \cdot 4 = 1$$

⋮

$$f_4(14) = 4 \cdot 14 = 11$$

ii)

$$f_7(0) = 7 \cdot 0 = 0$$

$$f_7(1) = 7 \cdot 1 = 7$$

$$f_7(14) = 7 \cdot 14 = 8$$

$$f_8(0)$$

,

,

,

(2)

$$f_{11}(0) = 11 \cdot 0 = 0$$

,

,

,

$$f_{13}(0) = 0$$

$$f_{13}(1) = 13 \cdot 1 = 13$$

$$f_{13}(2) = 13 \cdot 2 = 11$$

,

,

$$f_{14}(0) = 0$$

$$f_{14}(1) = 14 \cdot 1 = 14$$

$$f_{14}(2) = 14 \cdot 2 = 13$$

,

,

$\{f_1, f_2, f_4, f_7, f_8, f_{11}, f_{13}, f_{14}\} \in \text{Aut}(\mathbb{Z}_{15}) = \text{Aut}(\mathbb{Z}_5)$ dir

$$5A) D_5 = \{ \langle x, y \rangle \mid x^5 = y^2 = e \text{ ve } xy = yx^4 \}$$

$$= \{ e, x, x^2, x^3, x^4, y, xy, x^2y, x^3y, x^4y \}$$

$$o(D_5) = 10$$

$$o(yx^3) = ?$$

$$yx^3yx^3 = yx^2 \underline{xy} x^3 = yx^2 yx^{4+3} = yx^2 yx^2$$

$$= yx \underline{xy} x^2 = yx yx^{4+2}$$

$$= yx \underline{yx}$$

$$= y yx^{4+1} = y^2 x^5 = e$$

$$\Rightarrow o(yx^3) = 2 \text{ dir. } (5)$$

$$[x, yx^3] = ?$$

$$[x, yx^3] = x yx^3 x^{-1} (yx^3)^{-1} \text{ den}$$

$$= x yx^2 x^{-3} y^{-1} = \underline{xyx^{-1}} y^{-1} = \underline{yx^3} y^{-1}$$

$$= yx^3 y$$

$$= yx^2 y$$

$$= yx^2 yx^4$$

$$xy = yx^4 \Rightarrow xyx^{-1} = yx^3$$

$$(5) yx^4 x^{-1} y^{-1} = yx^3 y^{-1} = yx^3 y$$

$$\Rightarrow [x, yx^3] = x^2 \text{ bulurur.} = x^2 yy = x^2 \text{ bulurur.}$$

5b) $\mathbb{Z}_4 \times \mathbb{Z}_6$ $o(a, b) = 6$ olan elemanlar?

$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
1 4 2 4

(2)

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
1 6 3 2 3 6

(2)

$$o(a, b) = \text{lcm}(o(a), o(b)) = 6$$

$$o(a) = 1 \quad \vee \quad o(b) = 6 \Rightarrow (0, 1), (0, 5) \quad (2)$$

$$o(a) = 2 \quad \vee \quad o(b) = 3 \Rightarrow (2, 2), (2, 4) \quad (2)$$

$$o(a) = 2 \quad \vee \quad o(b) = 6 \Rightarrow (2, 1), (2, 5) \quad (2)$$

6 eleman vardır.

$$\{(0, 1), (0, 5), (2, 2), (2, 4), (2, 1), (2, 5)\}$$

mantıken 6 olan
elemanlar